

Experiment	<i>Lab 04: Experiment MWT – Forces and Moments on a Canard Delta Wing</i>
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Summary

This experiment's goal was to use a force balance in a closed-circuit wind tunnel to investigate the aerodynamic forces and moments operating on a canard-delta wing model. The wind used was the Embry-Riddle Micaplex Wind Tunnel. Lift, drag, side force, and pitching, rolling, and yawing moments were measured at different angles of attack during the experiment. For aerodynamic tares, data processing was employed to guarantee precise force readings. The experiment also looked into how yaw angle and Reynolds number affected aerodynamic properties at high angles of attack and low speeds. Plots of force and moment against the angle of attack were created using the data that was gathered.

At lower angles of attack, the lift coefficient (C_L) showed a nearly linear trend, but above 25 degrees, there were indications of flow separation. Pitch angle enhanced the drag coefficient (C_D), while higher Reynolds numbers indicated less drag because of better flow attachment. At low yaw angles, the side force, rolling moment, and yawing moment coefficients were nearly negligible, however, when a 10-degree yaw was applied, there were discernible changes. To evaluate aerodynamic efficiency, a lift-to-drag ratio ($\frac{C_L}{C_D}$ vs α) and a drag polar (C_L vs. C_D) were also examined. The findings showed improved aerodynamic performance was typically associated with higher Reynolds numbers. Higher Reynolds numbers resulted in stronger moments, and the pitching moment coefficient (C_m) rose with pitch angle.

Overall, the experimental findings showed how yaw and Reynolds number affect force and moment coefficients while also exhibiting the anticipated aerodynamic behavior.

Aim & Objectives

This lab focuses on developing a thorough grasp of a closed-circuit wind tunnel and measuring aerodynamic forces using a force balance. Additionally, it seeks to improve proficiency in assessing aerodynamic properties at high angles of attack and low speeds, as well as data correction utilizing aerodynamic tares.

- Recognize the parts and how a closed-circuit wind tunnel works.
- Learn how to quantify aerodynamic forces and moments using an external force balance.
- Use aerodynamic tares to adjust the data that was measured.
- Determine the pitching, rolling, and yawing moments as well as lift, drag, and side force.
- Examine how aerodynamics behaves at high angles of attack and low speeds.

Uncertainty Analysis

Ambient conditions and the uncertainties related to them are shown below:

Quantity	Measurement	Uncertainty
Pressure	102438 (Pa)	0.25 (Pa)
Temperature	295.15 (K)	0.5 (K)
Density	1.205 (kg/m ³)	0.5 (kg/m ³)

Table 1: Uncertainty measurements for basic quantities

Since the Micaplex Wind Tunnel was used for this experiment, we may assume that there is a 1% error in all of the force balance data.

Results & Discussion

Before beginning the data processing and analysis stage, the aerodynamic drag and interfering tares had to be removed. The MATLAB code can be found in the appendix.

Forces and Moments:

The initial step in this experiment was to create all of the force and moment graphs that the force balance generated. These include the following: yawing, rolling, pitching, drag, side force, and lift. Plotting this against the angle of attack will allow you to examine each event separately.

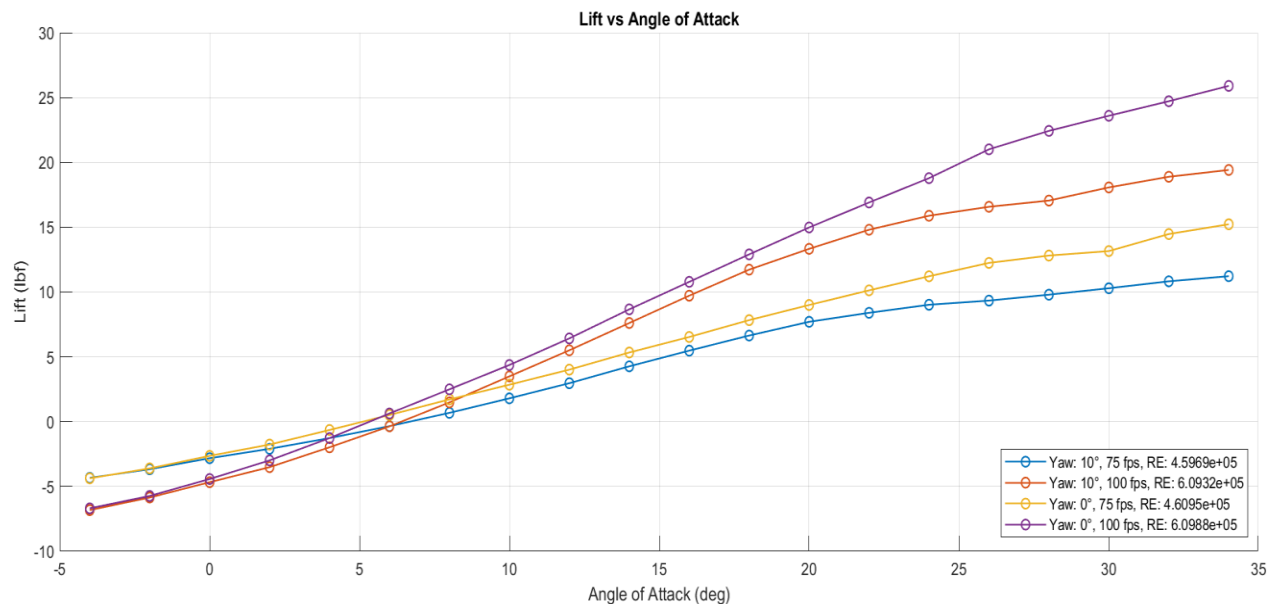


Figure 1: Lift vs AoA

This plot demonstrates the lift trend against the angle of attack quite clearly. It begins with a brief linear increase before pausing and continuing with a more gradual climb. Compared to 75 fps, the increased flow speed produces higher lift values. It seems that a yaw angle of 10 degrees has no influence on lift.

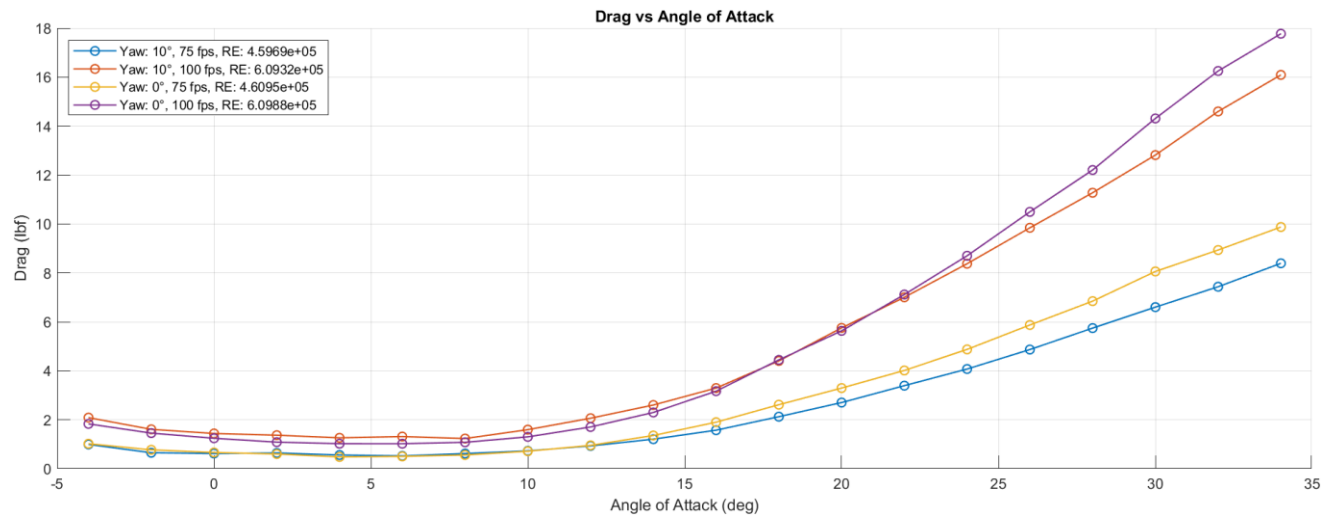


Figure 2: Drag vs AoA

According to the predicted pattern in aerodynamics, the graph above illustrates how the drag force rises as the pitch angle increases. At all angles, higher airspeed causes more drag than lower velocity. Furthermore, the drag is marginally increased by the 10-degree yaw angle.

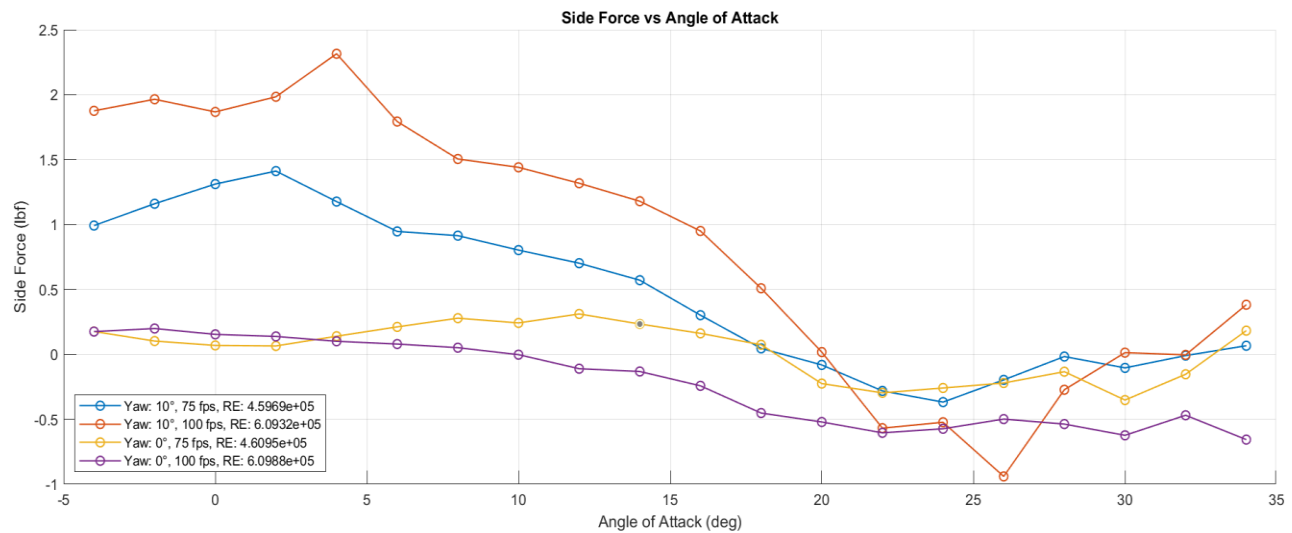


Figure 3: Side Force vs AoA

There are noticeable oscillations in this graph, which show how the side force changes with angle of attack. The side force is still quite modest at zero yaw, but at 10 degrees of yaw, it becomes stronger, particularly at lower pitch angles, and then it starts to decrease at higher angles. Moreover, adhering to the principle that the side forces have to be around the zero value.

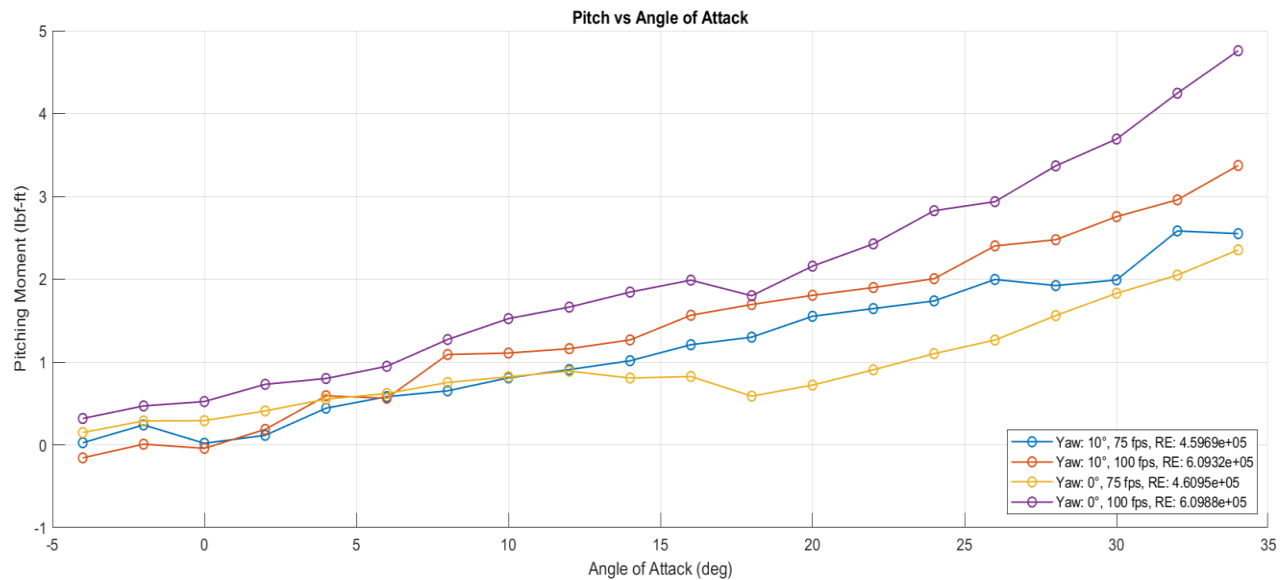


Figure 4: Pitching Moment vs AoA

The pitching moment versus angle of attack is displayed in the graph above. It is evident that the pitching moment rises as the pitch angle does. A stronger pitching moment is the result of the increased flow speed. Additionally, at higher pitch angles, the yaw angle becomes apparent.

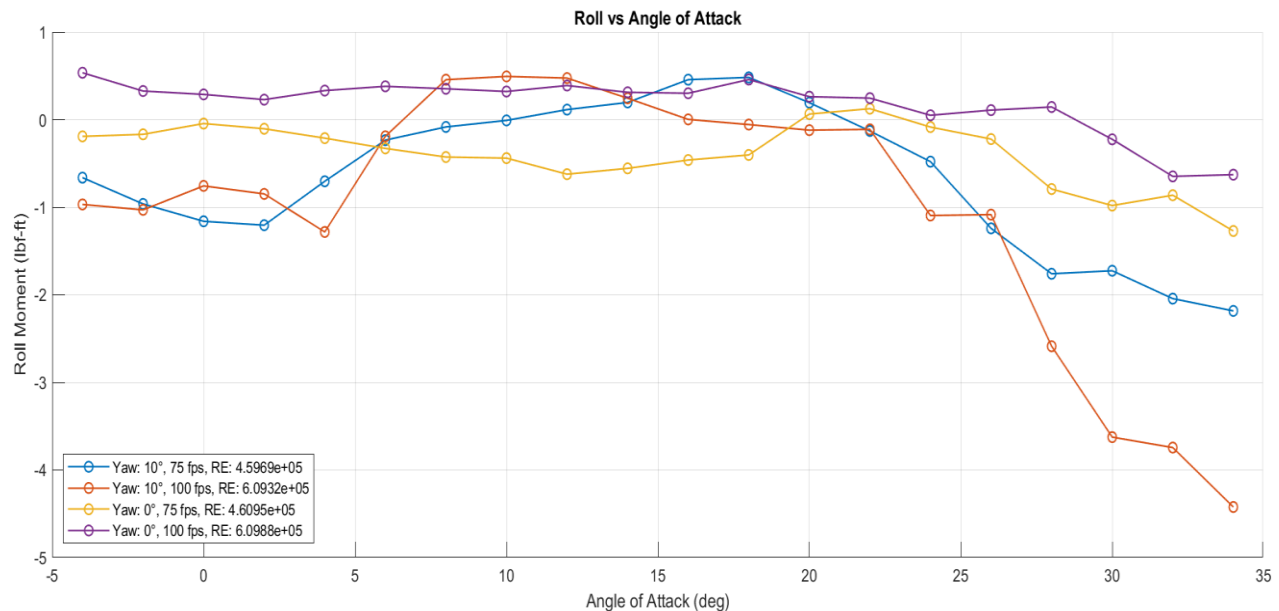


Figure 5: Rolling Moment vs AoA

The rolling moment versus angle of attack is displayed in the above figure. The rolling moment's trend is as it bounces around the 0 value before declining for extremely high pitch angles. Here, all speeds vary, but they all follow a similar pattern in relation to the yaw angle.

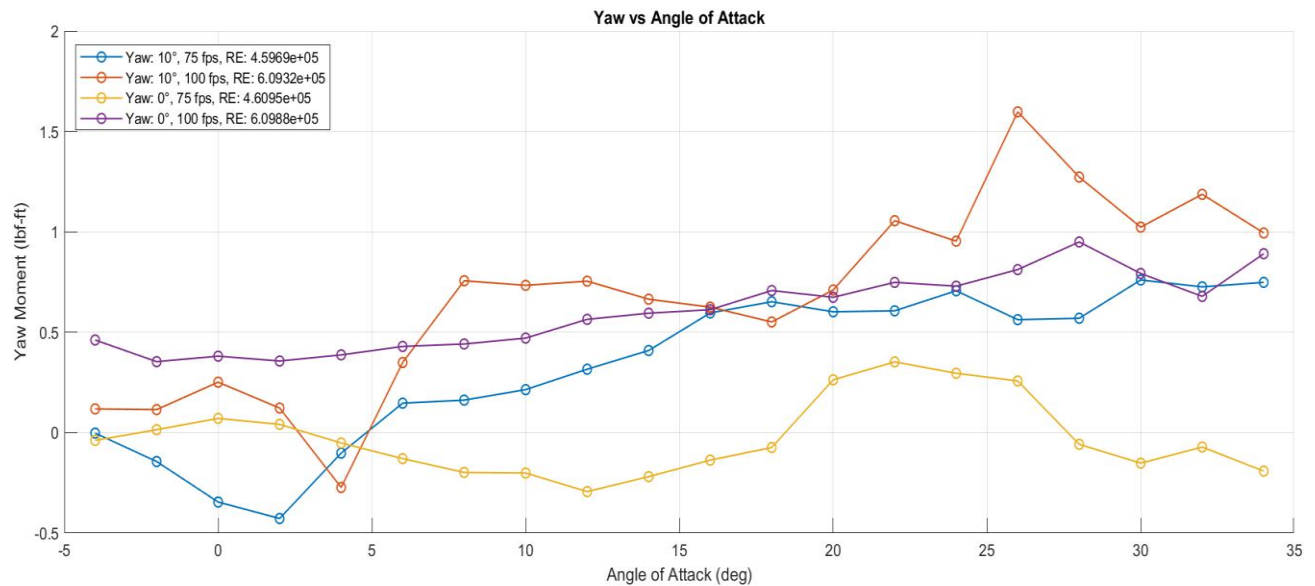


Figure 6: Yaw Moment vs AoA

This graph illustrates how the yawing moment varies greatly along the spectrum of AoA. once more exhibiting a distinct tendency to remain within the 0 values. As anticipated, a yaw angle of 10 degrees causes the yawing moment to increase, and a higher airspeed causes the yawing moment trend to become more noticeable.

Coefficients:

Plotting the lift and drag coefficients, C_L and C_D , for the canard model against α is the next step after analyzing the six graphs for forces and moments.

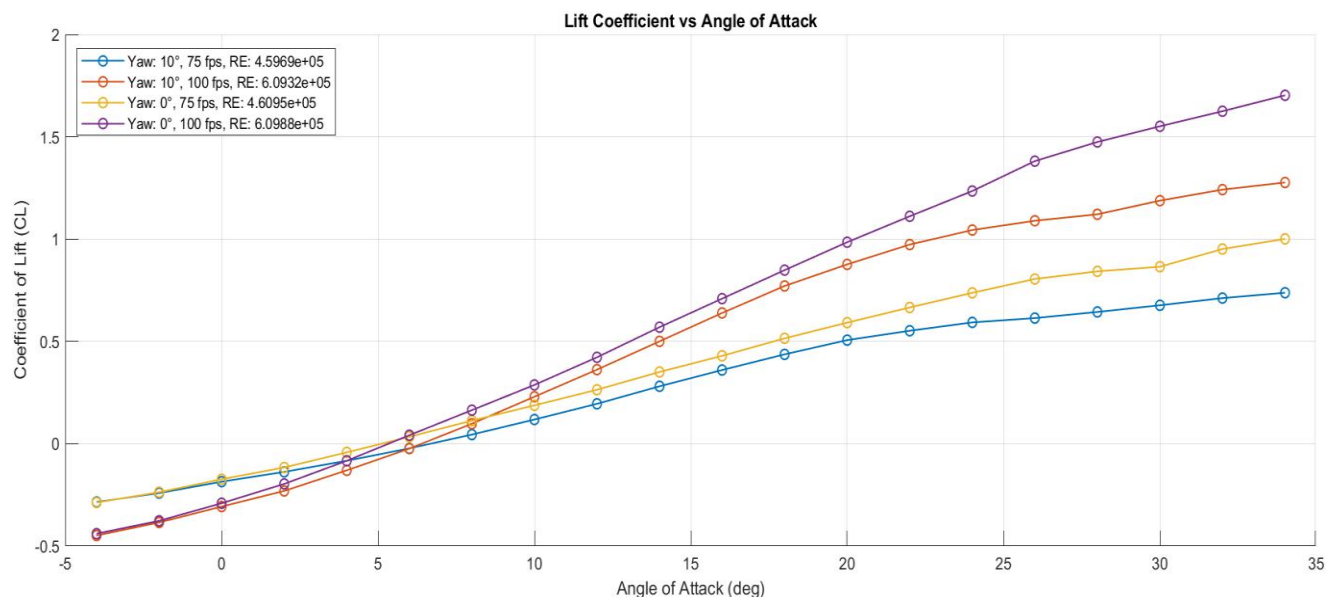


Figure 7: Lift Coefficient vs AoA

According to the predicted aerodynamic behavior, the graph above shows how the lift coefficient rises roughly linearly with pitch angle up to around 25 degrees. The lift coefficient then seems to peak at the angle of attack larger than 25 degrees and then level off slightly, indicating the start of flow separation over the lifting surfaces. At a given pitch angle, the lift coefficients produced by the higher Reynolds numbers, $Re: 6.0988 \times 10^5$ and 6.0932×10^5 are marginally higher than those produced by the lower Reynolds numbers.

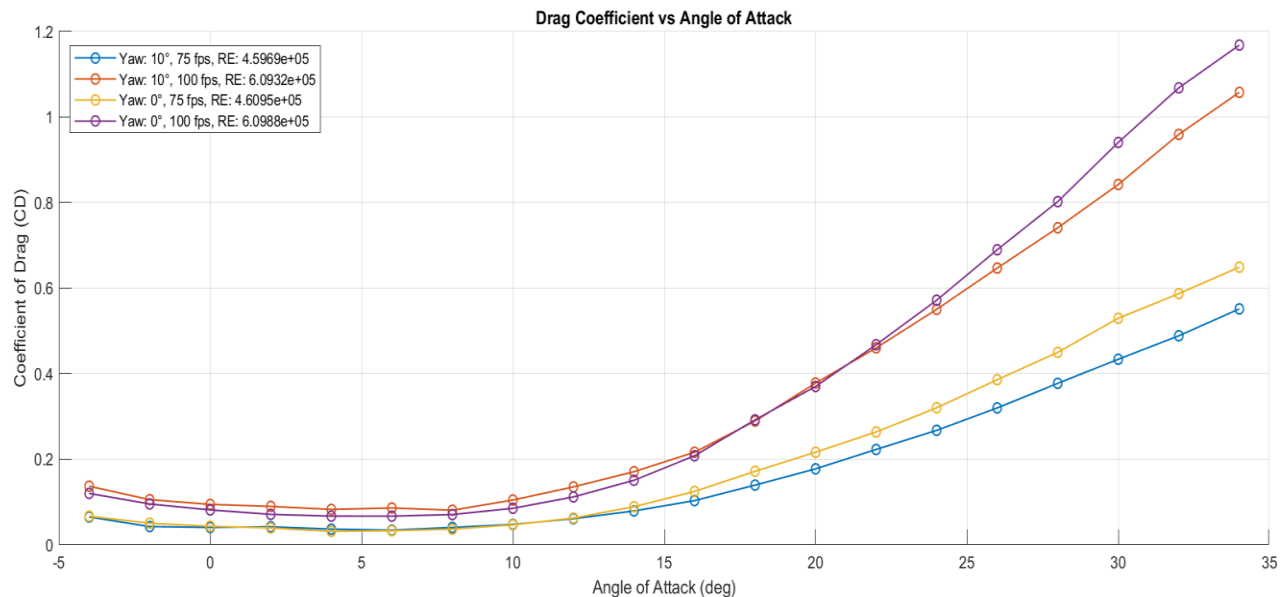


Figure 8: Drag Coefficient vs AoA

For modest angles of attack, the drag coefficient initially stays low and almost constant, as seen in the drag coefficient versus angle of attack graph. Then, C_D rises with an increase in AoA. Once more, flow separation and higher pressure drag happen when AoA is greater than 25°. Compared to situations with lower Reynolds numbers, those with higher Reynolds numbers exhibit a lower drag coefficient. Higher Reynolds numbers are suggested because they preserve greater flow attachment and lower total drag.

Overall:

We will now analyze the drag polar and lift-to-drag ratio.

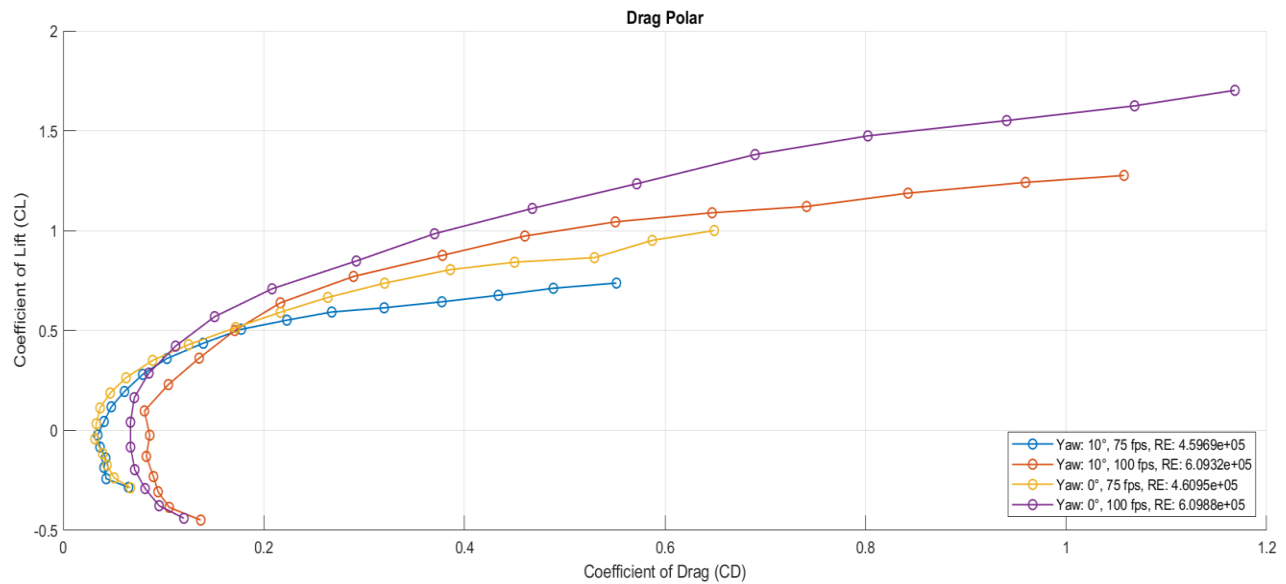


Figure 9: Drag Polar (CL vs CD)

The graph above illustrates a distinct pattern that begins when the coefficient of lift is zero and demonstrates how the coefficient of lift rises in tandem with an increase in the angle of attack. Although there is a noticeable change with the 10-degree yaw angle when the line separates from the 0-degree angle, all speeds follow the same general pattern.

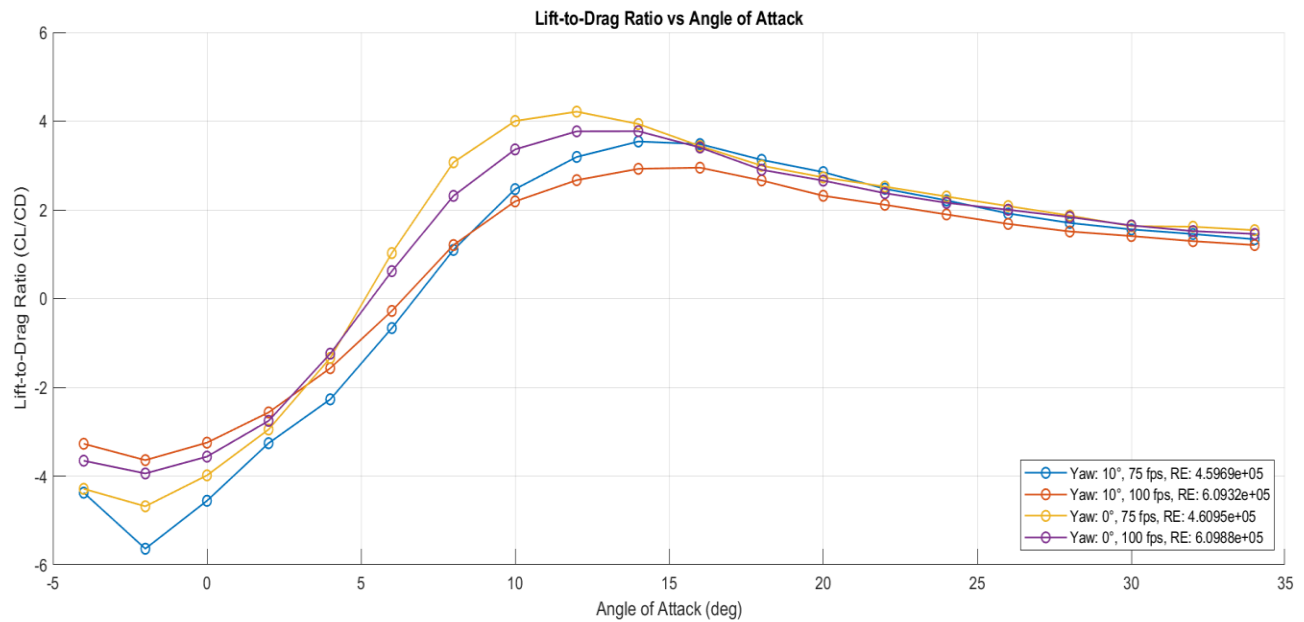


Figure 10: Lift-to-Drag ratio vs AoA

The model's ideal lift-to-drag ratio at each Reynolds number can be found using the graph above. A table is a more convenient way to present this data.

Reynolds	Speed (fps)	Yaw (degrees)	Best $\frac{C_L}{C_D}$	AoA (deg)
4.6095×10^5	75	0	4.21334	12
6.0988×10^5	100	0	3.76978	12
4.5969×10^5	75	10	3.19296	12
6.0932×10^5	100	10	2.66971	12

Table 2: Best CL/CD ratio

Moment & Force Coefficient:

Now, we will analyze the pitching moment coefficient for the model versus angle of attack.

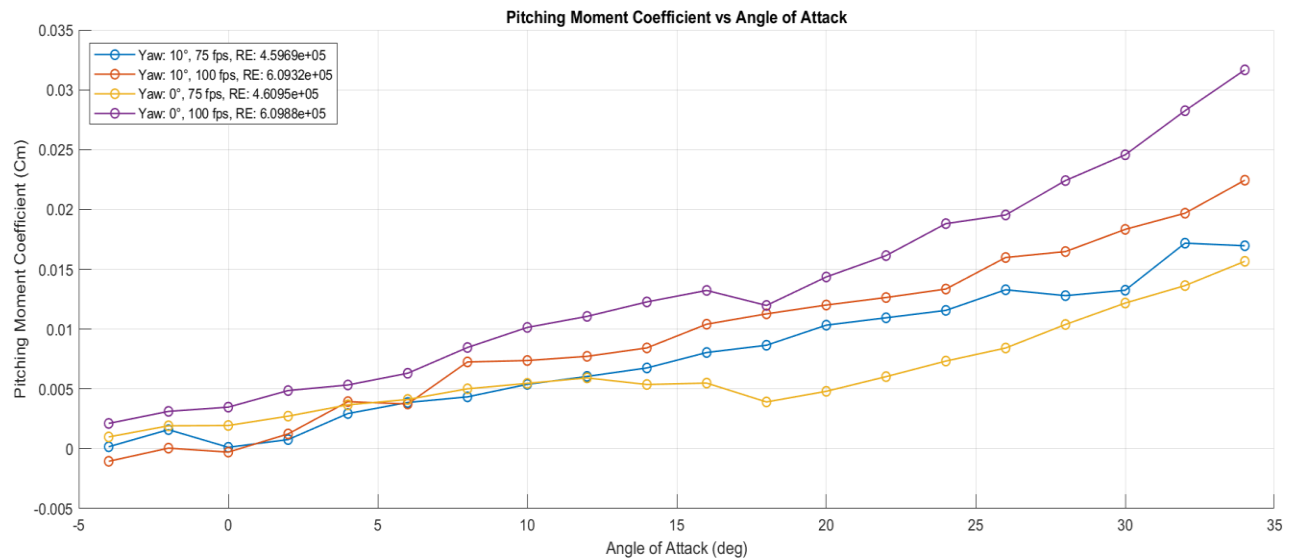


Figure 11: Cm vs AoA

The graph above clearly demonstrates a pattern. As the pitch angle rises, so does the pitching moment coefficient. This implies that the aerodynamic forces produce a higher nose-up moment as the angle of attack increases. At negative pitch angles, on the other hand, the pitching moment is close to zero or slightly negative, indicating a nose-down moment. Higher C_m values are typically associated with higher Reynolds numbers.

The increasing rate of C_m is lower at lower Reynolds numbers ($Re\ 4.5969 \times 10^5$ & 4.6095×10^5), indicating that the aerodynamic forces produce less nose-up moment at higher angles.

The following are the graphs for side force, rolling moment, and yaw moment coefficients versus angle of attack.

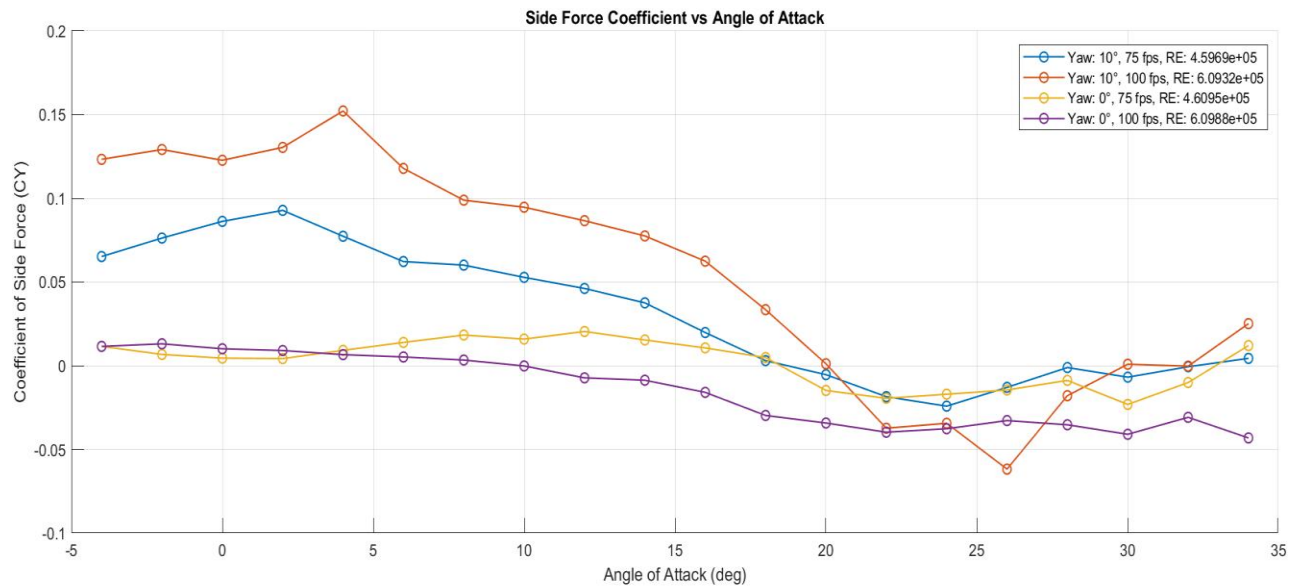


Figure 12: Side Force Coefficient vs AoA

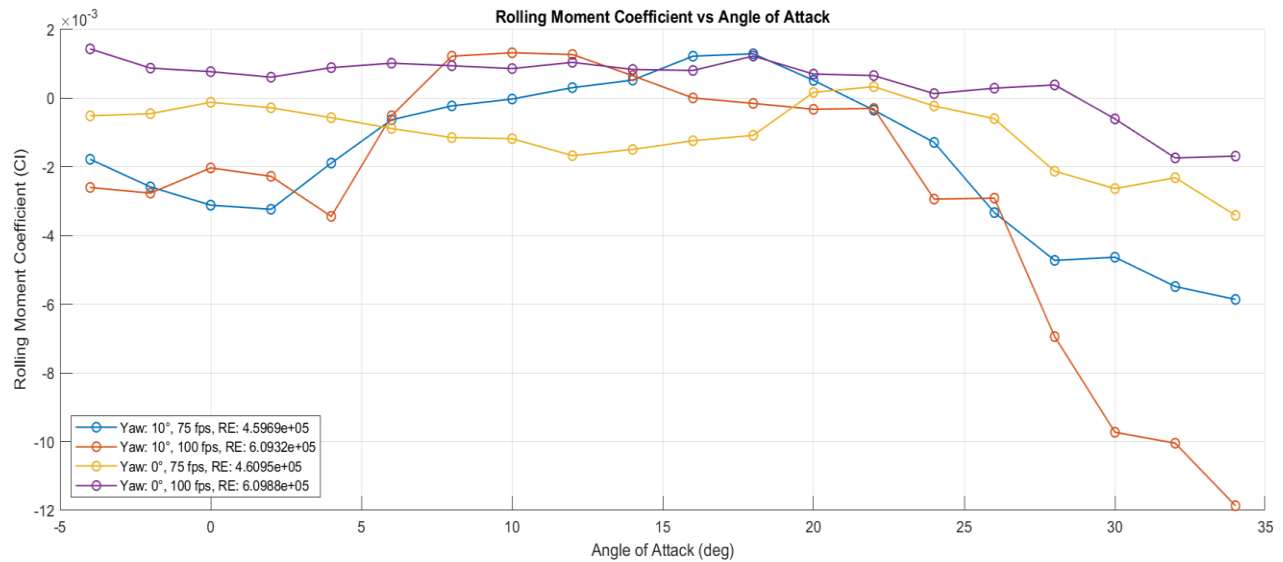


Figure 13: Rolling Moment Coefficient vs AoA

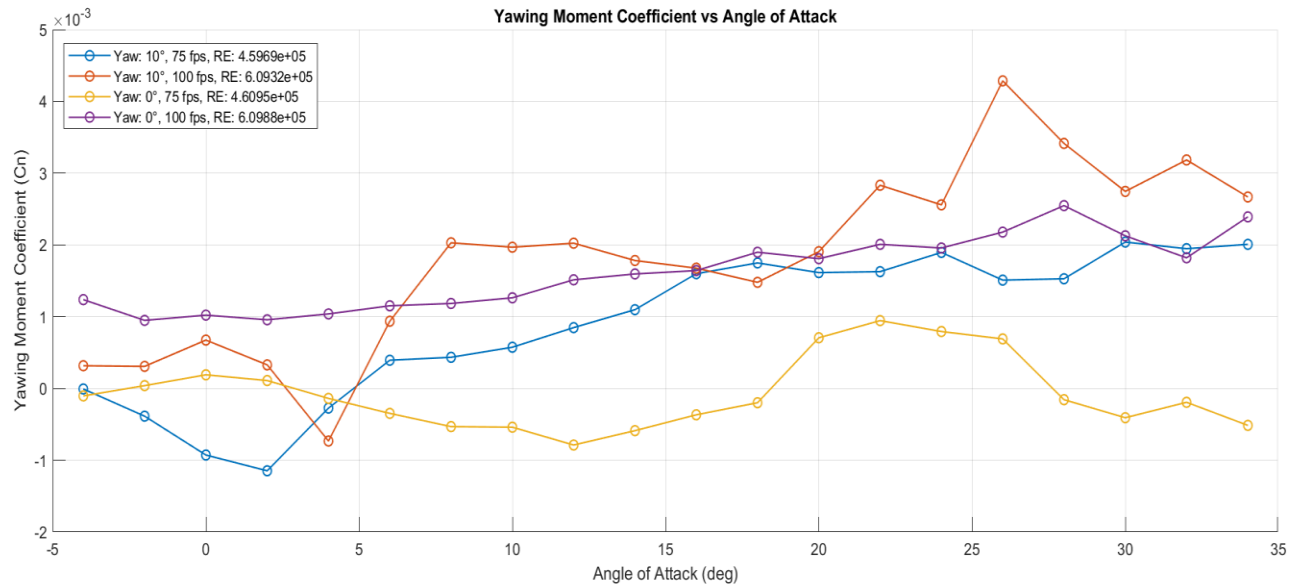


Figure 14: Yawing Moment Coefficient vs AoA

The side force coefficient (C_Y), rolling moment coefficient (C_l), and yawing moment coefficient (C_n) are plotted against angle of attack in the final three graphs. Since they all follow a similar pattern, they can all be examined collectively. Because of the scale, none of the points on any of the three graphs appear to deviate from one another by more than 0.1, indicating that they are all near to one another. This would be a fairly linear and steady line near the 0 value range when viewed on a bigger scale. In other words, neither the pitching moment nor the angle of attack significantly affect these coefficients.

References

1. Experiment MWT – Forces and Moments on a Canard Delta Wing (Micaplex Wind Tunnel): (2025). Embry-Riddle Aeronautical University.
2. AE315 Supplement – Experimental Aerodynamics Supplement: (2025). Embry-Riddle Aeronautical University.

Appendix

Appendix C: MATLAB Code

```
% Author: Anuranan Bharadwaj
% Lab 04: Experiment MWT
```

```
clc;
clear;
close all;
```

```
% Constants
```

```
R = 287;
Po = 102438;           % Pa
To = 295.15;           % K
rho0 = 1.205;           %Po / (R * To) kg/m^3
```

```
alpha = -4:2:34; % angle of attack from -4 to 34 at an increment of 2
```

```
%% Load data
```

```
%image
```

```
image10_75data = readtable␣
('MWT_DeltaCanardAeroTareData\Lab5_AeroTareData_Original\Image_Yaw10_75fps.csv',␣
'VariableNamingRule','preserve');
image10_100data = readtable␣
('MWT_DeltaCanardAeroTareData\Lab5_AeroTareData_Original\Image_Yaw10_100fps.csv',␣
'VariableNamingRule','preserve');
image0_75data = readtable␣
('MWT_DeltaCanardAeroTareData\Lab5_AeroTareData_Original\Image_Yaw0_75fps.csv',␣
'VariableNamingRule','preserve');
image0_100data = readtable␣
('MWT_DeltaCanardAeroTareData\Lab5_AeroTareData_Original\Image_Yaw0_100fps.csv',␣
'VariableNamingRule','preserve');
```

```
%invert
```

```
invert10_75data = readtable␣
('MWT_DeltaCanardAeroTareData\Lab5_AeroTareData_Original\Invert_Yaw10_75fps.csv',␣
'VariableNamingRule','preserve');
invert10_100data = readtable␣
('MWT_DeltaCanardAeroTareData\Lab5_AeroTareData_Original\invert_Yaw10_100fps.csv',␣
'VariableNamingRule','preserve');
invert0_75data = readtable␣
('MWT_DeltaCanardAeroTareData\Lab5_AeroTareData_Original\Invert_Yaw0_75fps.csv',␣
'VariableNamingRule','preserve');
invert0_100data = readtable␣
('MWT_DeltaCanardAeroTareData\Lab5_AeroTareData_Original\invert_yaw0_100fps.csv',␣
'VariableNamingRule','preserve');
```

```
%yaw
```

```
yaw10_75data = readtable␣
('MWT_raw_data_ALL\02DB_04DB\sec02\032125_084409_sec02_yaw10_75fps.csv',␣
'VariableNamingRule','preserve');
```

```

yaw10_100data = readtable␣
('MWT_raw_data_ALL\02DB_04DB\sec02\032125_085323_sec02_yaw10_100fps.csv',␣
'VariableNamingRule','preserve');
yaw0_75data = readtable␣
('MWT_raw_data_ALL\02DB_04DB\sec04\032125_095403_sec04_yaw0_75fps.csv',␣
'VariableNamingRule','preserve');
yaw0_100data = readtable␣
('MWT_raw_data_ALL\02DB_04DB\sec04\032125_100314_sec04_yaw0_100fps.csv',␣
'VariableNamingRule','preserve');

%% Analysis

%lift
lift10_75    = yaw10_75data.("WAFBC Lift")    - image10_75data.("WAFBC Lift") (2:2:end)␣
+ invert10_75data.("WAFBC Lift") (2:2:end);
lift10_100   = yaw10_100data.("WAFBC Lift")   - image10_100data.("WAFBC Lift") (2:2:end)␣
+ invert10_100data.("WAFBC Lift") (2:2:end);
lift0_75     = yaw0_75data.("WAFBC Lift")     - image0_75data.("WAFBC Lift") (2:2:end)␣
+ invert0_75data.("WAFBC Lift") (2:2:end);
lift0_100    = yaw0_100data.("WAFBC Lift")    - image0_100data.("WAFBC Lift") (2:2:end)␣
+ invert0_100data.("WAFBC Lift") (2:2:end);
lift = [lift10_75, lift10_100, lift0_75, lift0_100];

%drag
drag10_75    = yaw10_75data.("WAFBC Drag")    - image10_75data.("WAFBC Drag") (2:2:end)␣
+ invert10_75data.("WAFBC Drag") (2:2:end);
drag10_100   = yaw10_100data.("WAFBC Drag")   - image10_100data.("WAFBC Drag") (2:2:end)␣
+ invert10_100data.("WAFBC Drag") (2:2:end);
drag0_75     = yaw0_75data.("WAFBC Drag")     - image0_75data.("WAFBC Drag") (2:2:end)␣
+ invert0_75data.("WAFBC Drag") (2:2:end);
drag0_100    = yaw0_100data.("WAFBC Drag")    - image0_100data.("WAFBC Drag") (2:2:end)␣
+ invert0_100data.("WAFBC Drag") (2:2:end);
drag = [drag10_75, drag10_100, drag0_75, drag0_100];

%side force
sideForce10_75 = yaw10_75data.("WAFBC Side") - image10_75data.("WAFBC Side") (2:2:␣
end) + invert10_75data.("WAFBC Side") (2:2:end);
sideForce10_100 = yaw10_100data.("WAFBC Side") - image10_100data.("WAFBC Side") (2:␣
2:end) + invert10_100data.("WAFBC Side") (2:2:end);
sideForce0_75 = yaw0_75data.("WAFBC Side") - image0_75data.("WAFBC Side") (2:2:␣
end) + invert0_75data.("WAFBC Side") (2:2:end);
sideForce0_100 = yaw0_100data.("WAFBC Side") - image0_100data.("WAFBC Side") (2:2:␣
end) + invert0_100data.("WAFBC Side") (2:2:end);
side_force = [sideForce10_75, sideForce10_100, sideForce0_75, sideForce0_100];

%pitch
pitch10_75    = yaw10_75data.("WAFBC Pitch") - image10_75data.("WAFBC Pitch") (2:2:␣
end) + invert10_75data.("WAFBC Pitch") (2:2:end);
pitch10_100   = yaw10_100data.("WAFBC Pitch") - image10_100data.("WAFBC Pitch") (2:2:␣
end) + invert10_100data.("WAFBC Pitch") (2:2:end);

```

```

end)    + invert10_100data.("WAFBC Pitch") (2:2:end);
pitch0_75    = yaw0_75data.("WAFBC Pitch")    - image0_75data.("WAFBC Pitch") (2:2:
end)    + invert0_75data.("WAFBC Pitch") (2:2:end);
pitch0_100   = yaw0_100data.("WAFBC Pitch")    - image0_100data.("WAFBC Pitch") (2:2:
end)    + invert0_100data.("WAFBC Pitch") (2:2:end);
pitch = [pitch10_75, pitch10_100, pitch0_75, pitch0_100];

%yaw
yaw10_75    = yaw10_75data.("WAFBC Yaw")    - image10_75data.("WAFBC Yaw") (2:2:end)
+ invert10_75data.("WAFBC Yaw") (2:2:end);
yaw10_100   = yaw10_100data.("WAFBC Yaw")    - image10_100data.("WAFBC Yaw") (2:2:end)
+ invert10_100data.("WAFBC Yaw") (2:2:end);
yaw0_75     = yaw0_75data.("WAFBC Yaw")     - image0_75data.("WAFBC Yaw") (2:2:end)
+ invert0_75data.("WAFBC Yaw") (2:2:end);
yaw0_100    = yaw0_100data.("WAFBC Yaw")    - image0_100data.("WAFBC Yaw") (2:2:end)
+ invert0_100data.("WAFBC Yaw") (2:2:end);
yaw = [yaw10_75, yaw10_100, yaw0_75, yaw0_100];

%roll
roll10_75   = yaw10_75data.("WAFBC Roll")    - image10_75data.("WAFBC Roll") (2:2:end)
+ invert10_75data.("WAFBC Roll") (2:2:end);
roll10_100  = yaw10_100data.("WAFBC Roll")    - image10_100data.("WAFBC Roll") (2:2:end)
+ invert10_100data.("WAFBC Roll") (2:2:end);
roll0_75    = yaw0_75data.("WAFBC Roll")     - image0_75data.("WAFBC Roll") (2:2:end)
+ invert0_75data.("WAFBC Roll") (2:2:end);
roll0_100   = yaw0_100data.("WAFBC Roll")    - image0_100data.("WAFBC Roll") (2:2:end)
+ invert0_100data.("WAFBC Roll") (2:2:end);
roll = [roll10_75, roll10_100, roll0_75, roll0_100];

%main wing design (in/in^2)
bw  = 24.5;
crw = 16.01;
ctw = 3.75;
cw  = 9.88;
sw  = 242.01;

%canard design
bc  = 12.67;
crc = 9.69;
ctc = 1.67;
cc  = 5.68;
sc  = 71.97;

%reynolds' number calculation
%Reynolds Number per ft column and convert to main wing Reynolds number
RE10_75  = mean(yaw10_75data.("Reynolds Number per ft"));
RE10_100 = mean(yaw10_100data.("Reynolds Number per ft"));
RE0_75   = mean(yaw0_75data.("Reynolds Number per ft"));
RE0_100  = mean(yaw0_100data.("Reynolds Number per ft"));

```

```

%data sets
velocities = [
    mean(yaw10_75data.Velocity),    mean(yaw10_100data.Velocity), ...
    mean(yaw0_75data.Velocity),     mean(yaw0_100data.Velocity)
];
mean_velocity = mean(velocities);

%dynamic pressure
q = mean([
    mean(yaw10_75data.("Dynamic Pressure")),    mean(yaw10_100data.("Dynamic
Pressure")), ...
    mean(yaw0_75data.("Dynamic Pressure")),     mean(yaw0_100data.("Dynamic
Pressure"))
]);

%CL
CL = lift ./ (q * sw);

%CD
CD = drag ./ (q * sw);

%CY (side force)
CY = side_force ./ (q * sw);

%Cm
Cm = pitch ./ (q * sw * cw);

% Cl (roll)
Cl = roll ./ (q * sw * bw);

%Cn (yaw)
Cn = yaw ./ (q * sw * bw);

%drag Polar
%polyfit quadratic curve to obtain drag polar: CD = CD0 + K * CL^2
[p, S] = polyfit(CL, CD, 2);
CD0     = p(3);    % Parasitic drag coefficient
K       = p(2);    % Induced drag factor
R2 = 1 - (S.normr/norm(CD - mean(CD)))^2;

%% Plotting

%lift
figure(1)
set(figure(1), 'WindowStyle', 'Docked'); %docking
hold on
plot(alpha, lift10_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°,
75 fps, RE: %0.4e', RE10_75))

```

```

plot(alpha, lift10_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, lift0_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, lift0_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Lift (lbf)')
title('Lift vs Angle of Attack')
legend('Location', 'southeast')
hold off
grid on

%drag
figure(2)
set(figure(2), 'WindowStyle', 'Docked');
hold on
plot(alpha, drag10_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, drag10_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, drag0_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, drag0_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Drag (lbf)')
title('Drag vs Angle of Attack')
legend('Location', 'northwest')
hold off
grid on

%side force
figure(3)
set(figure(3), 'WindowStyle', 'Docked');
hold on
plot(alpha, sideForce10_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, sideForce10_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, sideForce0_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, sideForce0_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Side Force (lbf)')
title('Side Force vs Angle of Attack')
legend('Location', 'southwest')
hold off

```

```
grid on
```

```
%pitching moment
```

```
figure(4)
set(gcf,'WindowStyle','Docked');
hold on
plot(alpha, pitch10_75, 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75↵
fps, RE: %0.4e', RE10_75))
plot(alpha, pitch10_100, 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100↵
fps, RE: %0.4e', RE10_100))
plot(alpha, pitch0_75, 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps,↵
RE: %0.4e', RE0_75))
plot(alpha, pitch0_100, 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100↵
fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Pitching Moment (lbf-ft)')
title('Pitch vs Angle of Attack')
legend('Location', 'southeast')
hold off
grid on
```

```
%yawing moment
```

```
figure(5)
set(gcf,'WindowStyle','Docked');
hold on
plot(alpha, yaw10_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75↵
fps, RE: %0.4e', RE10_75))
plot(alpha, yaw10_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°,↵
100 fps, RE: %0.4e', RE10_100))
plot(alpha, yaw0_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75↵
fps, RE: %0.4e', RE0_75))
plot(alpha, yaw0_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100↵
fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Yaw Moment (lbf-ft)')
title('Yaw vs Angle of Attack')
legend('Location', 'northwest')
hold off
grid on
```

```
%rolling moment
```

```
figure(6)
set(gcf,'WindowStyle','Docked');
hold on
plot(alpha, roll10_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°,↵
75 fps, RE: %0.4e', RE10_75))
plot(alpha, roll10_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°,↵
100 fps, RE: %0.4e', RE10_100))
plot(alpha, roll0_75(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75↵
```



```

fps, RE: %0.4e', RE0_75))
plot(alpha, roll0_100(:,1), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°',
100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Roll Moment (lbf-ft)')
title('Roll vs Angle of Attack')
legend('Location', 'southwest')
hold off
grid on

%CL vs AoA
figure(7)
set(gcf, 'WindowStyle', 'Docked');
hold on
plot(alpha, CL(1:length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw:
10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, CL(length(alpha)+1:2*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName',
sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, CL(2*length(alpha)+1:3*length(alpha)), 'o-', 'LineWidth', 1,
'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, CL(3*length(alpha)+1:end), 'o-', 'LineWidth', 1, 'DisplayName', sprintf
('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Lift (CL)')
title('Lift Coefficient vs Angle of Attack')
legend('Location', 'northwest')
grid on
hold off

%CD vs AoA
figure(8)
set(gcf, 'WindowStyle', 'Docked');
hold on
plot(alpha, CD(1:length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw:
10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, CD(length(alpha)+1:2*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName',
sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, CD(2*length(alpha)+1:3*length(alpha)), 'o-', 'LineWidth', 1,
'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, CD(3*length(alpha)+1:end), 'o-', 'LineWidth', 1, 'DisplayName', sprintf
('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Drag (CD)')
title('Drag Coefficient vs Angle of Attack')
legend('Location', 'northwest')
grid on
hold off

%CY vs AoA

```

```

figure(9)
set(gcf,'WindowStyle','Docked');
hold on
plot(alpha, CY(1:length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, CY(length(alpha)+1:2*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, CY(2*length(alpha)+1:3*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, CY(3*length(alpha)+1:end), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Coefficient of Side Force (CY)')
title('Side Force Coefficient vs Angle of Attack')
legend('Location', 'northeast')
grid on
hold off

```

%Cm vs AoA

```

figure(10)
set(gcf,'WindowStyle','Docked');
hold on
plot(alpha, Cm(1:length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, Cm(length(alpha)+1:2*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, Cm(2*length(alpha)+1:3*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, Cm(3*length(alpha)+1:end), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Pitching Moment Coefficient (Cm)')
title('Pitching Moment Coefficient vs Angle of Attack')
legend('Location', 'northwest')
grid on
hold off

```

%Cl vs AoA

```

figure(11)
set(gcf,'WindowStyle','Docked');
hold on
plot(alpha, Cl(1:length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, Cl(length(alpha)+1:2*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, Cl(2*length(alpha)+1:3*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, Cl(3*length(alpha)+1:end), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))

```

```

xlabel('Angle of Attack (deg)')
ylabel('Rolling Moment Coefficient (Cl)')
title('Rolling Moment Coefficient vs Angle of Attack')
legend('Location', 'southwest')
grid on
hold off

% Cn vs AoA
figure(12)
set(figure(12), 'WindowStyle', 'Docked');
hold on
plot(alpha, Cn(1:length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, Cn(length(alpha)+1:2*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(alpha, Cn(2*length(alpha)+1:3*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(alpha, Cn(3*length(alpha)+1:end), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Yawing Moment Coefficient (Cn)')
title('Yawing Moment Coefficient vs Angle of Attack')
legend('Location', 'northwest')
grid on
hold off

% CL vs CD - drag polar
figure(13)
set(figure(13), 'WindowStyle', 'Docked');
hold on
plot(CD(1:length(alpha)), CL(1:length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 75 fps, RE: %0.4e', RE10_75))
plot(CD(length(alpha)+1:2*length(alpha)), CL(length(alpha)+1:2*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e', RE10_100))
plot(CD(2*length(alpha)+1:3*length(alpha)), CL(2*length(alpha)+1:3*length(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e', RE0_75))
plot(CD(3*length(alpha)+1:end), CL(3*length(alpha)+1:end), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Coefficient of Drag (CD)')
ylabel('Coefficient of Lift (CL)')
title('Drag Polar')
legend('Location', 'southeast')
grid on
hold off

% CL/CD vs AoA
figure(14)
set(figure(14), 'WindowStyle', 'Docked');
hold on

```

```
plot(alpha, CL(1:length(alpha))./CD(1:length(alpha)), 'o-', 'LineWidth', 1,↵
'DisplayName', sprintf('Yaw: 10°, 75 fps, RE: %0.4e', RE10_75))
plot(alpha, CL(length(alpha)+1:2*length(alpha))./CD(length(alpha)+1:2*length(alpha)), ↵
'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 10°, 100 fps, RE: %0.4e',↵
RE10_100))
plot(alpha, CL(2*length(alpha)+1:3*length(alpha))./CD(2*length(alpha)+1:3*length ↵
(alpha)), 'o-', 'LineWidth', 1, 'DisplayName', sprintf('Yaw: 0°, 75 fps, RE: %0.4e',↵
RE0_75))
plot(alpha, CL(3*length(alpha)+1:end)./CD(3*length(alpha)+1:end), 'o-', 'LineWidth', ↵
1, 'DisplayName', sprintf('Yaw: 0°, 100 fps, RE: %0.4e', RE0_100))
xlabel('Angle of Attack (deg)')
ylabel('Lift-to-Drag Ratio (CL/CD)')
title('Lift-to-Drag Ratio vs Angle of Attack')
legend('Location', 'southeast')
grid on
hold off
```